

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Story Idea Input	Enter a one-line idea in a web form Select genre and number of panels
FR-2	Outline Generation	Create the story outline with Gemini Flash Return title and panel summaries
FR-3	Story Writing	Write the full story with the second Gemini model Split the story into per-panel captions
FR-4	Panel Image Generation	Build an image prompt for each panel Generate images with Stable Diffusion v1.5
FR-5	Comic Layout & Preview	Arrange panels and captions on a page Show live progress and the final comic
FR-6	PDF Export	Export the finished comic as a PDF with fpdf2 Download the file from the browser

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Single-page web interface; a user needs only an idea, genre and panel count to start
NFR-2	Security	API keys are kept in environment variables and never exposed to the browser; inputs are validated
NFR-3	Reliability	Automatic retry on Gemini rate-limit (429) errors; a second model is used when quota is exhausted
NFR-4	Performance	About 12.4 s per panel image and about 104 s end to end on Apple M5 (16 GB, MPS fp16)
NFR-5	Availability	Runs locally and on demand; image generation works offline once the model is downloaded
NFR-6	Scalability	Stateless FastAPI routes; panel count is configurable; models can be swapped or moved to a GPU server